Given a quasi-Banach function space E, for $0 < r < \infty$, $E^{(r)}$ will denote the quasi-Banach function space defined by $E^{(r)} = \{f : |f|^r \in E\}$ equipped with the quasi-norm $\|f\|_{E^{(r)}} = \big\||f|^r\big\|_E^{\frac{1}{r}}$. Note that

$$p_{E^{(r)}} = rp_E, \quad q_{E^{(r)}} = rq_E, \tag{3.2}$$

and if $0 < p, q < \infty$, then

$$(E^{(p)})^{(q)} = E^{(pq)}. \tag{3.3}$$

If E is a symmetric Banach function space and $p \geq 1$, then $E^{(p)}$ is a symmetric Banach function space.

Let E_i be a quasi-Banach function space for $i = 1, 2$. The pointwise product space $E_1 \odot E_2$ is defined by

$$E_1 \odot E_2 = \{f \in L_0(0, \infty) : f = f_1 f_2, f_i \in E_i, i = 1, 2\}$$

with functional $\|\cdot\|_{E_1 \odot E_2}$ being defined by

$$\|f\|_{E_1 \odot E_2} = \inf\{\|f\|_{E_1}\|f\|_{E_2} : f = f_1 f_2, f_i \in E_i, i = 1, 2\}.$$

We need the following lemmas (see [**17**, Theorem 1 (iii), Corollary 2] and [**18**, Theorem 6]).

Lemma 3.1. *Let E and F be two symmetric Banach function spaces.*
(i) *If $0 < p < \infty$, then $(E \odot F)^{(p)} = E^{(p)} \odot F^{(p)}$.*
(ii) *$L_1(0, \infty) = E \odot E^{\times}$.*
(iii) *If $1 < p < \infty$, then $(E^{(p)})^{\times} = (E^{\times})^{(p)} \odot L_{p'}(0, \infty)$.*

Lemma 3.2. *Let E be a symmetric Banach function space which is separable or has the Fatou property with $p_E > p$. Then $E^{(\frac{1}{p})}$ can be renormed as a symmetric Banach space.*

Proof. We first consider the case of $0 < p \leq 1$. Since E is symmetric and $\frac{1}{p} \geq 1$, we have that $E^{(\frac{1}{p})}$ is a symmetric Banach function space (see [**17**, p 53]). For $p > 1$, by [**6**, Theorem 3.2], we have that E is an interpolation space for the couple $\big(L_p(0, \infty), L_\infty(0, \infty)\big)$. It follows that $E^{(\frac{1}{p})}$ is an interpolation space for the couple $\big(L_1(0, \infty), L_\infty(0, \infty)\big)$ (see [**6**, Theorem 3.5]). Thus according to Lemma 2.2 in [**3**], we get that $E^{(\frac{1}{p})}$ can be renormed as a symmetric Banach function space. $\square$

3.2. Noncommutative symmetric spaces and martingales. For a given noncommutative measure space $(\mathcal{M}, \tau)$ and a symmetric quasi-Banach function space $(E, \|\cdot\|_E)$ on $(0, \infty)$, we define the corresponding noncommutative symmetric space by setting

$$E(\mathcal{M}, \tau) = \{x \in L_0(\mathcal{M}) : \mu_t(x) \in E\}$$

equipped with the quasi-norm

$$\|x\|_{E(\mathcal{M},\tau)} := \|\mu_t(x)\|_E.$$

It is well-known that $E(\mathcal{M}, \tau)$ (denoted by $E(\mathcal{M})$ for convenience) is a quasi-Banach space. Note that if $0 < p < \infty$ and $E = L_p(0, \infty)$, then $E(\mathcal{M}, \tau) = L_p(\mathcal{M}, \tau)$ is the usual noncommutative L_p-space associated with $(\mathcal{M}, \tau)$. We refer to [**1, 3**] for more details and historical references on these spaces.

Let $(\mathcal{M}, \tau)$ be a noncommutative measure admitting a martingale structure, that is, there exists an increasing sequence of von Neumann subalgebras $(\mathcal{M}_n)_{n\geq 1}$

of $\mathcal{M}$ such that the union of the $\mathcal{M}_n$'s is weak*-dense in $\mathcal{M}$. For each $n \geq 1$, the unique conditional expectation $\mathcal{E}_n$ from $\mathcal{M}$ onto $\mathcal{M}_n$ extends to a contractive projection from $L_p(\mathcal{M}, \tau)$ onto $L_p(\mathcal{M}_n, \tau_n)$ for all $1 \leq p \leq \infty$. More generally, if E is a symmetric Banach function space on $(0, \infty)$ that belongs to $\mathrm{Int}(L_1, L_\infty)$—the interpolation spaces, then $\mathcal{E}_n$ is bounded from $E(\mathcal{M}, \tau)$ onto $E(\mathcal{M}_n, \tau_n)$.

A sequence $x = (x_n)_{n\geq 1}$ is called a $E(\mathcal{M})$-martingale if $x_n \in E(\mathcal{M})$ and $\mathcal{E}_n(x_{n+1}) = x_n$, $\forall n \geq 1$. In this case, we set $\|x\|_E = \sup_{n\geq 1} \|x_n\|_E$. If $\|x\|_E < \infty$, then x is called a bounded $E(\mathcal{M})$-martingale. The column and row conditioned Hardy spaces $\mathsf{h}_E^c(\mathcal{M})$ and $\mathsf{h}_E^r(\mathcal{M})$ are defined to be respectively the completions of the space of all the finite $E(\mathcal{M})$-martingales under the associated quasi-norms $\|x\|_{\mathsf{h}_E^c} = \|s_c(x)\|_E$ and $\|x\|_{\mathsf{h}_E^r} = \|s_r(x)\|_E$, where $s_c(x)$ and $s_r(x)$ are the column and row conditioned square functions of x, defined by (with the convention that $\mathcal{E}_0 = \mathcal{E}_1$)

$$s_c(x) = (\sum_{k\geq 1} \mathcal{E}_{k-1}|dx_k|^2)^{1/2} \quad \text{and} \quad s_r(x) = s_c(x^*).$$

In general, we have no explicit description of elements in $\mathsf{h}_E^c(\mathcal{M})$ or $\mathsf{h}_E^r(\mathcal{M})$, that is, not all the elements of $\mathsf{h}_E^c(\mathcal{M})$ and $\mathsf{h}_E^r(\mathcal{M})$ can be represented by a martingale. However, if $E = L_p(0, \infty)$ for $0 < p < \infty$, then $\mathsf{h}_E^c(\mathcal{M}) = \mathsf{h}_p^c(\mathcal{M})$ and $\mathsf{h}_E^r(\mathcal{M}) = \mathsf{h}_p^r(\mathcal{M})$, namely the column and row conditioned Hardy spaces of noncommutative martingales. As remarked in [**28**], if E is a symmetric Banach function space on $(0, \infty)$ with the Fatou property which is an interpolation space for the couple $\big(L_p(0, \infty), L_q(0, \infty)\big)$ for some $1 < p < q < \infty$, then every element of $\mathsf{h}_E^c(\mathcal{M})$ and $\mathsf{h}_E^r(\mathcal{M})$ can be represented by a martingale.

3.3. Symmetric space moment inequality. Starting with the p-moment inequalities—(2.13), we are able to obtain the moment characterisation in terms of symmetric spaces in a more direct way. In the $\mathsf{bmo}^c(\mathcal{M})$ case, our approach is much more efficient than the interpolation arguments given in [**3**].

We first introduce the symmetric Lipschitz spaces $\Lambda_{\beta,E}^c(\mathcal{M})$ and $\Lambda_{\beta,E}^r(\mathcal{M})$.

Definition 3.3. Let $\beta \geq 0$. Let E be a separable symmetric Banach function space on $(0, \infty)$ with $1 < p_E \leq q_E < \infty$. We define

$$\Lambda_{\beta,E}^c(\mathcal{M}) = \{x \in L_2(\mathcal{M}) : \|x\|_{\Lambda_{\beta,E}^c} < \infty\},$$

where

$$\|x\|_{\Lambda_{\beta,E}^c} = \max\{\|\mathcal{E}_1(x)\|_\infty, \ \sup_n \sup_{e\in\mathcal{P}(\mathcal{M}_n)} (\tau(e))^{-\beta} \|(x - x_n)\frac{e}{\|e\|_E}\|_{\mathsf{h}_E^c}\}$$

and

$$\Lambda_{\beta,E}^r(\mathcal{M}) = \{x \in L_2(\mathcal{M}) : x^* \in \Lambda_{\beta,E}^c(\mathcal{M})\}.$$

Let $\Lambda_{\beta,E}^{0,c}(\mathcal{M})$ and $\Lambda_{\beta,E}^{0,r}(\mathcal{M})$ be their subspaces of all x with $\mathcal{E}_1(x) = 0$.

Theorem 3.4. *Let $\beta \geq 0$ and $x \in \Lambda_\beta^c(\mathcal{M})$. Let E be a separable symmetric Banach function space on $(0, \infty)$ with $1 < p_E \leq q_E < \infty$. Then we have that*

$$\alpha_E^{-1}\|x\|_{\Lambda_\beta^c} \leq \|x\|_{\Lambda_{\beta,E}^c} \leq \beta_E\|x\|_{\Lambda_\beta^c}.$$

The constants α_E and β_E depend on E. The same inequalities hold for $\|\cdot\|_{\Lambda_\beta^r}$ and $\|\cdot\|_{\Lambda_{\beta,E}^r}$.

LEMMA 3.5. *Let E be a separable symmetric Banach function space on $(0,\infty)$ with $1 < p_E \le q_E < \infty$. Choose p such that $q_E < p$. Then we have that $E = F^{(p')} \odot L_p(0,\infty)$, where $F = (E^{\times(\frac{1}{p'})})^{\times}$. Here and below, for $1 \le p \le \infty$, p' denotes the conjugate index of p.*

PROOF. By (3.1), it follows that $1 < p' < p_{E^\times} \le q_{E^\times} < \infty$. Since $E^\times$ is symmetric and has the Fatou property, by Lemma 2.2, $E^{\times(\frac{1}{p'})}$ can be renormed as a symmetric Banach function space. Then by (3.3) and (iii) of Lemma 3.1, we have that

$$E = (E^\times)^\times = ([E^{\times(\frac{1}{p'})}]^{(p')})^\times = ([E^{\times(\frac{1}{p'})}]^\times)^{(p')} \odot L_p(0,\infty) = F^{(p')} \odot L_p(0,\infty).$$

□

We will use repeatedly the following fact which follows from Theorem 2 of [**19**].

LEMMA 3.6. *Let E and F be two symmetric Banach function spaces on $(0,\infty)$. Then $E\odot F$ is a symmetric quasi-Banach function space on $(0,\infty)$ and the following formula holds:*

$$\|\chi_{[0,t]}\|_{E\odot F} = \|\chi_{[0,t]}\|_E \|\chi_{[0,t]}\|_F \quad \textit{for} \quad t \in (0,\infty).$$

PROOF OF THEOREM 3.4. We may assume $\mathcal{E}_1(x) = 0$. Let F, p and p' be as in Lemma 3.5. Fix n and $P \in \mathcal{P}(\mathcal{M}_n)$. By definition $\mu_t(P) = \chi_{[0,\tau(P)]}(t)$, and thus by Lemma 3.5 and Lemma 3.6, we have

$$\|P\|_E = \|P\|_{F^{(p')}} \|P\|_p.$$

Therefore, by the equality $E^{\frac12} = F^{(\frac{p'}{2})} \odot L_{\frac{p}{2}}(0,\infty)$, we have that

$$\begin{aligned}
(\tau(P))^{-\beta}\|(x-x_n)\frac{P}{\|P\|_E}\|_{\mathsf{h}^c_E} &= (\tau(P))^{-\beta}\Big\|\frac{P}{\|P\|_E}s_c^2(x-x_n)\frac{P}{\|P\|_E}\Big\|^{\frac12}_{E^{(\frac12)}} \\
&\le (\tau(P))^{-\beta}\Big\|\frac{P}{\|P\|_E^2}\Big\|^{\frac12}_{F^{(\frac{p'}{2})}}\|Ps_c^2(x-x_n)P\|^{\frac12}_{\frac{p}{2}} \\
&= (\tau(P))^{-\beta}\frac{1}{(\tau(P))^{\frac1p}}\|(x-x_n)P\|_{\mathsf{h}^c_p} \qquad (3.4)
\end{aligned}$$

which implies that $\|x\|_{\Lambda^c_{\beta,E}} \le \|x\|_{\Lambda^c_{\beta,p}}$. Then by (iii) of Theorem 2.10, we have that $\|x\|_{\Lambda^c_{\beta,E}} \le \beta_E\|x\|_{\Lambda^c_\beta}$.

Conversely, choose p such that $p < p_E$. Then by Lemma 3.2, $E^{(\frac1p)}$ can be renormed as a symmetric Banach function space. Thus by (ii) of Lemma 3.1, we have that $L_1(0,\infty) = E^{(\frac1p)} \odot E^{(\frac1p)\times}$. It follows that $L_p(0,\infty) = E \odot F$ by (i) of Lemma 3.1, where $F = (E^{(\frac1p)\times})^{(p)}$. Hence, for all $e \in \mathcal{P}(\mathcal{M}_n)$, by Lemma 3.6, we have

$$(\tau(e))^{\frac1p} = \|e\|_E\|e\|_F. \qquad (3.5)$$

Using the equality $L_{\frac{p}{2}}(0,\infty) = E^{(\frac12)} \odot F^{(\frac12)}$, one has

$$\begin{aligned}
\|(x-x_n)\frac{e}{(\tau(e))^{\frac1p}}\|_{\mathsf{h}^c_p} &= \Big\|\frac{e}{\|e\|_F}\frac{e}{\|e\|_E}s_c^2(x-x_n)\frac{e}{\|e\|_E}\frac{e}{\|e\|_F}\Big\|^{\frac12}_{\frac{p}{2}} \\
&\le \Big\|\frac{e}{\|e\|_F^2}\Big\|^{\frac12}_{F^{(\frac12)}}\Big\|\frac{e}{\|e\|_E}s_c^2(x-x_n)\frac{e}{\|e\|_E}\Big\|^{\frac12}_{E^{(\frac12)}} \\
&= \Big\|(x-x_n)\frac{e}{\|e\|_E}\Big\|_{\mathsf{h}^c_E} \qquad (3.6)
\end{aligned}$$

which implies $\|x\|_{\Lambda^c_{\beta,p}} \leq \|x\|_{\Lambda^c_{\beta,E}}$. Therefore, by Theorem 2.10, we get the desired result

$$\|x\|_{\Lambda^c_\beta} \leq \alpha_E \|x\|_{\Lambda^c_{\beta,E}}.$$

$\square$

REMARK 3.7. When $0 < p_E \leq q_E \leq 1$, it is easy to obtain the left inequality from the proof of Theorem 3.4.

4. The atomic Hardy space $\mathsf{h}_{p,E}(\mathcal{M})$ for $0 < p \leq 1$

In this section, we show that the noncommutative Hardy space $\mathsf{h}^c_{p,E}(\mathcal{M})$ $(0 < p \leq 1)$ defined via symmetric space atoms is also a predual space of Lipshitz space $\Lambda^c_{\frac{1}{p}-1}(\mathcal{M})$.

DEFINITION 4.1. Let $0 < p \leq 1$. Let E be a separable symmetric Banach function space on $(0,\infty)$ and $1 < p_E \leq q_E < \infty$. An element $a \in L_p(\mathcal{M})$ is called a $(p,E)_c$-atom with respect to $(\mathcal{M}_n)_{n\geq 1}$, if there exist $n \geq 1$ and a projection $e \in \mathcal{M}_n$ such that:
(i) $\mathcal{E}_n(a) = 0$;
(ii) $r(a) \leq e$;
(iii) $\|a\|_{\mathsf{h}^c_E} \leq (\tau(e))^{1-1/p}\|e\|^{-1}_{E^\times}$.
Replacing (ii) by $l(a) \leq e$, we have the notion of a $(p,E)_r$-atom.

Note that if $E = L_q(0,\infty)$ for $1 < q < \infty$, they are (p,q)-atoms defined in [**4**, Definition 4.1].

DEFINITION 4.2. Let $0 < p \leq 1$. Let E be a separable symmetric Banach function space on $(0,\infty)$ and $1 < p_E \leq q_E < \infty$. We define $\mathsf{h}^c_{p,E}(\mathcal{M})$ as the space of all operators $x \in L_p(\mathcal{M})$ which admits a decomposition

$$x = \sum_k \lambda_k a_k,$$

where for each k, a_k is either a $(p,E)_c$-atom or an element of the unit ball of $L_p(\mathcal{M}_1)$, and $\lambda_k \in \mathbb{C}$ satisfying $\sum_k |\lambda_k|^p < \infty$. We equip this space with the p-norm:

$$\|x\|_{\mathsf{h}^c_{p,E}} = \inf \,(\sum_k |\lambda_k|^p)^{\frac{1}{p}},$$

where the infimum is taken over all decompositions of x described above. We also define the subspace:

$$\mathsf{h}^{0,c}_{p,E}(\mathcal{M}) = \{x \in \mathsf{h}^c_{p,E}(\mathcal{M}) :\ \mathcal{E}_1(x) = 0\}.$$

Similarly, we define $\mathsf{h}^r_{p,E}(\mathcal{M})$ and $\mathsf{h}^{0,r}_{p,E}(\mathcal{M})$.

LEMMA 4.3. *Let $0 < p \leq 1$. If a is a $(p,E)_c$-atom, then*

$$\|a\|_{\mathsf{h}^c_p} \leq 1.$$

The similar inequality holds for $(p,E)_r$-atom.

PROOF. Let e be a projection associated with a satisfying (i)-(iii) of Definition 4.1. Then $s_c(a) = es_c(a) = s_c(a)e$ (see [**2**, Proposition2.2]). Nothing that $p < p_E$, similar with the proof of (3.6),

$$\begin{aligned}
\|a\|_{\mathsf{h}_p^c} &\leq \|s_c(a)\|_{E^{(\frac{1}{2})}}^{\frac{1}{2}} \|e\|_F \\
&= \|a\|_{\mathsf{h}_E^c} \frac{(\tau(e))^{\frac{1}{p}}}{\|e\|_E} \text{ (by (3.5))} \\
&\leq \frac{\tau(e)}{\|e\|_{E^\times}\|e\|_E} \text{ (by (iii) of Definition 4.1)} \\
&= 1 \quad \text{(by } L_1(0,\infty) = E \odot E^\times \text{ and Lemma 3.6).}
\end{aligned}$$

Thus we obtain the desired result. □

LEMMA 4.4. *Let E be a separable symmetric Banach function space on $(0,\infty)$ and $1 < p_E \leq q_E < \infty$. For a given $q_E < q_0 < \infty$ and $q = \max\{2, q_0\}$, $L_q(\mathcal{M})$ embeds densely and continuously into $\mathsf{h}_{p,E}^c(\mathcal{M})$.*

PROOF. Let $x \in L_q(\mathcal{M})$. We decompose it as a linear combination of two atoms:

$$x = C_q C_E \|x - \mathcal{E}_1(x)\|_q \frac{x - \mathcal{E}_1(x)}{C_q C_E \|x - \mathcal{E}_1(x)\|_q} + \|\mathcal{E}_1(x)\|_q \frac{\mathcal{E}_1(x)}{\|\mathcal{E}_1(x)\|_q},$$

where C_E is the constant in the inequality $\|y\|_E \leq C_E \|y\|_q$ for all $y \in L_q(\mathcal{M})$, and C_q is the constant in the noncommutative Burkholder inequality $\|y\|_{\mathsf{h}_q^c} \leq C_q \|y\|_q$ for all $y \in L_q(\mathcal{M})$ (see [**15**]). Then $\mathcal{E}_1(x)/\|\mathcal{E}_1(x)\|_q \in L_q(\mathcal{M}_1) \subset L_1(\mathcal{M}_1)$ and

$$\Big\|\frac{\mathcal{E}_1(x)}{\|\mathcal{E}_1(x)\|_q}\Big\|_1 = \frac{\|\mathcal{E}_1(x)\|_1}{\|\mathcal{E}_1(x)\|_q} \leq 1.$$

Also,

$$\frac{x - \mathcal{E}_1(x)}{C_q C_E \|x - \mathcal{E}_1(x)\|_q} = \frac{x - \mathcal{E}_1(x)}{C_q C_E \|x - \mathcal{E}_1(x)\|_q} \cdot 1 \doteq ae.$$

Clearly, $\mathcal{E}_1(a) = 0$ and

$$\Big\|\frac{x - \mathcal{E}_1(x)}{C_q C_E \|x - \mathcal{E}_1(x)\|_q}\Big\|_{\mathsf{h}_E^c} \leq \frac{C_E \|x - \mathcal{E}_1(x)\|_{\mathsf{h}_q^c}}{C_q C_E \|x - \mathcal{E}_1(x)\|_q} \leq 1.$$

Thus

$$\|x\|_{\mathsf{h}_{p,E}^c} \leq C_q C_E \|x - \mathcal{E}_1(x)\|_q + \|\mathcal{E}_1(x)\|_q \leq (2C_q C_E + 1)\|x\|_q.$$

The density is trivial. □

PROPOSITION 4.5. *Let $0 < p \leq 1$ and $\beta = 1/p - 1$. Let E be a separable symmetric Banach function space on $(0,\infty)$. If $1 < p_E \leq q_E < \infty$, then $\big(\mathsf{h}_{p,E}^{0,c}(\mathcal{M})\big)^* = \Lambda_{\beta,E^\times}^{0,c}(\mathcal{M})$ with equivalent norms. More precisely,*

(i) *Every $y \in \Lambda_{\beta,E^\times}^{0,c}(\mathcal{M})$ defines a continuous linear functional on $\mathsf{h}_{p,E}^{0,c}(\mathcal{M})$ by*

$$\varphi_y(x) = \tau(y^* x) \tag{4.1}$$

for all $x \in \mathsf{h}_{p,E}^{0,c}(\mathcal{M})$, and $\|\varphi_y\|_{\left(\mathsf{h}_{p,E}^{0,c}(\mathcal{M})\right)^*} \leq \|y\|_{\Lambda_{\beta,E^\times}^{0,c}}$;

(ii) *Conversely, each $\varphi \in \big(\mathsf{h}_{p,E}^{0,c}(\mathcal{M})\big)^*$ is given as (4.1) by some $y \in \Lambda_{\beta,E^\times}^{0,c}(\mathcal{M})$, and $\|y\|_{\Lambda_{\beta,E^\times}^{0,c}} \leq C_E \|\varphi_y\|_{\left(\mathsf{h}_{p,E}^{0,c}(\mathcal{M})\right)^*}$.*

Similarly, $\big(\mathsf{h}_{p,E}^{0,r}(\mathcal{M})\big)^* = \Lambda_{\beta,E^\times}^{0,r}(\mathcal{M})$ with equivalent norms.

PROOF. (i) Let $y \in \Lambda^{0,c}_{\beta,E^\times}(\mathcal{M})$. If a is a $(p,E)_c$-atom with $\mathcal{E}_n(a)=0$ for some $n \geq 1$ and $a = ae$ for some projection $e \in \mathcal{M}_n$ satisfying $\|a\|_{\mathsf{h}^c_E} \leq (\tau(e))^{1-1/p}\|e\|^{-1}_{E^\times}$, then using the duality inclusion $\mathsf{h}^c_{E^\times}(\mathcal{M}) \subset \big(\mathsf{h}^c_E(\mathcal{M})\big)^*$ with constant 1 (see [**3**], Theorem 2.5]), we have that

$$\begin{aligned}
|\tau(y^*a)| &= |\tau\big((y-y_n)^*ae\big)| \\
&\leq \|(y-y_n)e\|_{\mathsf{h}^c_{E^\times}}\|a\|_{\mathsf{h}^c_E} \\
&\leq \|(y-y_n)e\|_{\mathsf{h}^c_{E^\times}}(\tau(e))^{1-1/p}\|e\|^{-1}_{E^\times} \\
&\leq \|y\|_{\Lambda^{0,c}_{\beta,E^\times}}.
\end{aligned}$$

(ii) Let $\varphi \in \big(\mathsf{h}^{0,c}_{p,E}(\mathcal{M})\big)^*$. Set $q = \max\{q_0, 2\}$, where $q_E < q_0 < \infty$. By Lemma 4.4, we can find $y \in L_{q'}(\mathcal{M})$ such that

$$\varphi(x) = \tau(y^*x), \quad \forall x \in L_q(\mathcal{M}).$$

Fix $n \geq 1$ and $e \in \mathcal{P}(\mathcal{M}_n)$. For a fixed arbitrary and small enough $\varepsilon > 0$, by the inclusion $\big(\mathsf{h}^c_E(\mathcal{M})\big)^* \subset \mathsf{h}^c_{E^\times}(\mathcal{M})$ with contant C_E, we may choose $x \in L_q(\mathcal{M})$ such that $\|x\|_{\mathsf{h}^c_E} \leq 1$ so that

$$C_E|\tau(e(y-y_n)^*x)| + \varepsilon \geq \|(y-y_n)e\|_{\mathsf{h}^c_{E^\times}}.$$

Clearly, we may assume that $\mathcal{E}_n(x) = 0$ and $xe = x$. Set

$$a = \frac{x}{\|x\|_{\mathsf{h}^c_E}(\tau(e))^{1/p-1}\|e\|_{E^\times}}.$$

Then a is a $(p,E)_c$-atom and

$$\begin{aligned}
\|\varphi\| &\geq |\tau\big((y-y_n)^*a\big)| \\
&= \frac{1}{\|x\|_{\mathsf{h}^c_E}(\tau(e))^{1/p-1}\|e\|_{E^\times}}|\tau(e(y-y_n)^*x)| \\
&\geq \frac{1}{C_E(\tau(e))^{\beta}\|e\|_{E^\times}}(\|(y-y_n)e\|_{\mathsf{h}^c_{E^\times}} - \varepsilon).
\end{aligned}$$

By the arbitrariness of ε, and taking supremum over n and $e \in \mathcal{P}(\mathcal{M}_n)$, we get $C_E\|\varphi\| \geq \|y\|_{\Lambda^{0,c}_{\beta,E^\times}}$. □

By Proposition 4.5 and Theorem 3.4, we arrive at the main result of this subsection.

THEOREM 4.6. *Let $0 < p \leq 1$ and $\beta = 1/p-1$. Let E be a separable symmetric Banach function space on $(0,\infty)$. If $1 < p_E \leq q_E < \infty$, then $\big(\mathrm{h}^{0,c}_{p,E}(\mathcal{M})\big)^* = \Lambda^{0,c}_\beta(\mathcal{M})$ with equivalent norms. The similar results hold for the row and mixture spaces.*

REMARK 4.7. By Lemma 4.3, we have the obvious inclusion $\mathsf{h}^c_{p,E}(\mathcal{M}) \subset \mathsf{h}^c_p(\mathcal{M})$ for $0 < p \leq 1$. When $p = 1$, the converse inclusion is also true [**3**]; it is, however, an open question in the cases $0 < p < 1$ even though they have the same dual from Theorem 4.6 and [**4**, Theorem 5.4].

REMARK 4.8. We may consider the crude symmetric atoms. Let $0 < p \leq 1$. Let E be a separable symmetric Banach function space on $(0,\infty)$ with $1 < p_E \leq q_E < \infty$. An element $a \in L_p(\mathcal{M})$ is called a $(p,E)_c$-crude atom with respect to $(\mathcal{M}_n)_{n\geq 1}$, if there exist $n \geq 1$ and a factorization $a = yb$ such that: